DIRECTIONS *for questions 1 to 5:* Answer these questions on the basis of the information given below.

Fresla is a grocery store that has exactly six employees – Busk, Dusk, Fusk, Gusk, Husk and Lusk. The monthly salaries of the six employees are $347, $476, $538, $627, $836, and $968, not necessarily in the same order. In addition to the monthly salary, the employees may also be paid a bonus, which can be paid at any time during the year. The *monthly income* of any employee for a month is defined as the sum of the monthly salary of that employee for that month and any bonus that may have been received in that month. Further, the *annual income* of any employee is defined as the sum of the *monthly incomes* paid to the employee during the twelve months of the year, i.e., from January to December.

In September of a particular year, Elmond Rusk, the owner of Fresla, decided that he would give a bonus to each of the six employees such that the annual income of each employee would become an integral multiple of $1000. Elmond Rusk paid the bonus to all the six employees in the month of September itself, ensuring that each employee was paid the least possible bonus. No other bonus was paid to any employee during the year.

It is also known that

i. the bonus paid to Dusk was the same as the monthly salary of Busk; the bonus paid to Gusk was the same as the monthly salary of Fusk.

ii. the difference between the monthly salary and the bonus was not the least for Lusk, while the *monthly income* in September of Busk was less than the monthly salary of either Husk or Gusk.

Q1. **DIRECTIONS** *for questions 1 to 5:* Select the correct alternative from the given choices.
What was the *annual income* of Gusk?

a) **$5000**

b) **$8000**

c) **$11000**

d) **Either $11000 or $5000**

Q2. **DIRECTIONS** *for questions 1 to 5:* Select the correct alternative from the given choices.
What is the difference between the bonus paid to Dusk and the monthly salary of Fusk?

- a) **$492**
- b) **$360**
- c) **Either $360 or $492**
- d) **Either $360 or $621**

Q3. **DIRECTIONS** *for questions 1 to 5:* Select the correct alternative from the given choices.
Which of the following statements is true regarding the bonus paid to Lusk?

- a) **It is less than the bonus paid to Fusk.**
- b) **It is less than the monthly salary of Dusk.**
- c) **It is the same as the monthly salary of Gusk.**
- d) **It is greater than the *monthly income* of Husk for September.**

Q4. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.
The *monthly income* for September of which of the six persons is a multiple of 11?

- a) **Fusk**
- b) **Gusk**
- c) **Either Dusk or Gusk**
- d) **Dusk**

Q5. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.

For which employee is their *monthly income* for September as a proportion of their *annual income* the highest?

- a) **Husk**
- b) **Lusk**
- c) **Dusk**
- d) **Fusk**

DIRECTIONS *for questions 6 to 10:* Answer the questions on the basis of the information given below.
All the students of Class X of a school wrote four exams, each in a different subject among Maths, Physics, Chemistry and Biology. Three students, Ram, Lalit and Kishore, are from this class.

Chart-1 below provides the marks scored by each of Ram, Lalit and Kishore in each subject, while Chart-2 provides the number of students in the class who scored more than each of the three students in each subject. It is known that, in any exam, no two students received the same mark.

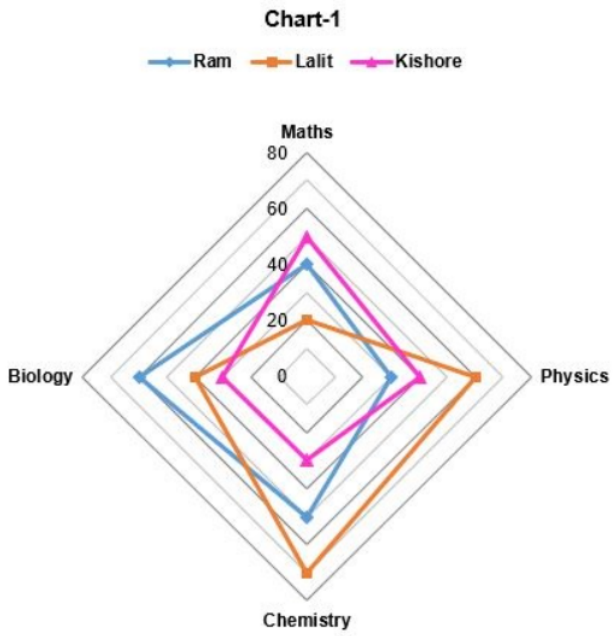
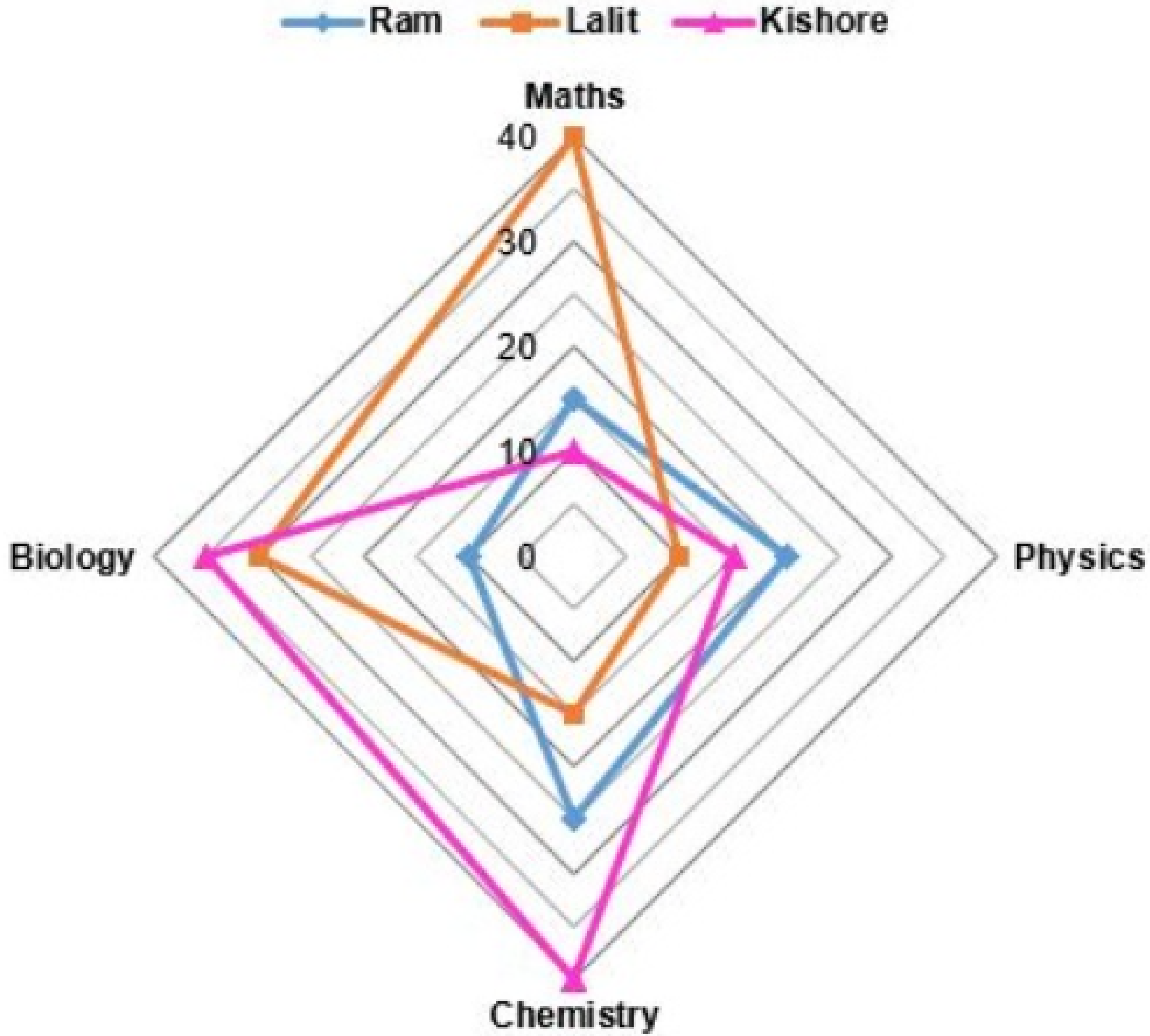


Chart-2



Ram
Lalit
Kishore
Maths
40
30
20
10
0
Biology
Physics
Chemistry

Q6. DIRECTIONS *for question 6:* Select the correct alternative from the given choices.
How many students scored more than 30 but less than 60 in Physics?

- a) **10** ✗ **Your answer is incorrect**
- b) **9**
- c) **20**
- d) **19**

Q7. DIRECTIONS *for question 7:* Type in your answer in the input box provided below the question.

What is the minimum number of students in the class?

Q8. DIRECTIONS *for question 8:* Select the correct alternative from the given choices.

If in exactly *n* subjects, the mark that any student received is a positive integer, what is the maximum possible value of *n*?

- a) **4**
- b) **3**
- c) **2**
- d) **1**

Q9. DIRECTIONS *for questions 9 and 10:* Type in your answer in the input box provided below the question.

If the marks received by any student in one of the subjects is a positive integer, what is the maximum possible number of students in the class?

Q10. DIRECTIONS *for questions 9 and 10:* Type in your answer in the input box provided below the question.

If the number of students who scored more than Lalit in one of the subjects is the same as the number of students who scored less than Kishore in another subject, what is the maximum number of students in the class?

DIRECTIONS *for questions 11 to 15:* Answer the questions on the basis of the information given below.

On each day, Harish, the warden of a hostel, decides the subjects to be read by each of the 194 students in his hostel on that day. He assigns one or more subjects among Civics, History and Economics to each student.

On a particular day, Day 1, 40 students were assigned only History, 20 students were assigned only Economics and 24 students were assigned only Civics. The number of students who were assigned only History and Economics was twice the number of students who were assigned all the three subjects which, in turn, was twice the number of students who were assigned only Economics and Civics. The number of students who were assigned History was 141.

On Day 2 and Day 3, Harish assigned the subjects in the following manner:

- Half the students who were assigned only History the previous day were additionally assigned Economics but not Civics.
- Half the students who were assigned only Economics the previous day were additionally assigned Civics but not History.
- Half the students who were assigned only Civics the previous day were additionally assigned both History and Economics.
- One third of the students who were assigned all the three subjects the previous day were assigned only History; one-third were assigned only Economics; one-third were assigned only Civics.

Q11. **DIRECTIONS** *for questions 11 to 13:* Type in your answer in the input box provided below the question.
How many students were assigned Economics on Day 2?

Q12. **DIRECTIONS** *for questions 11 to 13:* Type in your answer in the input box provided below the question.

How many students were assigned Economics and History on Day 3?

Q13. **DIRECTIONS** *for questions 11 to 13:* Type in your answer in the input box provided below the question.
What is the difference between the number of students who were assigned only History on Day 3 and the number of students who were assigned Civics on Day 2?

Q14. **DIRECTIONS** *for questions 14 and 15:* Select the correct alternative from the given choices.
On which of the given days were the number of students who were assigned Civics less than that who were assigned Economics?

- a) **Only Day 3**
- b) **Both Day 2 and Day 3**
- c) **Both Day 1 and Day 2**
- d) **None of Day 1, Day 2 or Day 3**

Q15. DIRECTIONS *for questions 14 and 15:* Select the correct alternative from the given choices.
What is the maximum number of students who would have been assigned only History on all the three days?

- a) **10**
- b) **13**
- c) **20**
- d) **7**

DIRECTIONS *for questions 16 to 20:* Answer the questions on the basis of the information given below.

In an archery tournament, each player gets exactly one attempt to hit the target in each round of the tournament. The tournament comprises n rounds in total. In any round, if a player hits the target, points are awarded, and if he misses the target, points are deducted. The manner in which the points are awarded/deducted is given below:

Awarding of points: For the first round in which a player hits the target, he is awarded three points. For every consecutive round that he continues to hit the target, the number of points awarded to him for the round increases by 1. If he misses the target in any round(s) and then hits the target in the successive round, only three points are awarded for the round in which he hits the target, and from this round onwards, for every consecutive round that he continues to hit the target, the number of points awarded to him for the round again increases by 1. The points are awarded in a similar manner for all the rounds in which the player hits the target.

Deducting of Points: For the first round in which a player misses the target, he is penalized one point (i.e., one point is deducted). For every consecutive round that he continues to miss the target, the number of points that he is penalized for the round increases by 1. If he hits the target in any round(s) and then misses the target in the successive round, he is penalized only one point for the round in which he misses the target, and from this round onwards, for every consecutive round that he continues to miss the target, the number of points that he is penalized for the round again increases by 1. The player is penalized in a similar manner for all the rounds in which he misses the target.

Q16. **DIRECTIONS** *for questions 16 to 20:* Select the correct alternative from the given choices.
If n = 10, and the player missed the target in exactly three rounds, what is the minimum possible score of the player?

a) **21**

b) **23**

c) **19**

d) **36** ✗ Your answer is incorrect

Q17. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.
If n = 10 and the player hit the target in exactly two rounds, what is the minimum possible score of the player?

a) **–28**

b) **–29** ✗ **Your answer is incorrect**

c) **–30**

d) **None of the above**

Q18. **DIRECTIONS** *for questions 16 to 20:* Select the correct alternative from the given choices.
If n = 10 and the player scored exactly 26 points, what is the maximum number of rounds in which he missed the target?

- a) **3**
- b) **5**
- c) **6**
- d) **4**

Q19. **DIRECTIONS** *for questions 16 to 20:* Select the correct alternative from the given choices.
If n = 60 and the player hit the target in exactly four rounds, what is the maximum possible score of the player?

- a) **–1578**
- b) **–408**
- c) **–330**
- d) **–264**

Q20. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.
It is known that n = 95 and the player hit the target in exactly five rounds. If the score of the player was the maximum possible, in which of the following rounds must he have hit the target?

- a) **19^{th} round**
- b) **80^{th} round**
- c) **20^{th} round**
- d) **Either 19^{th} round or 20^{th} round**